

DHARSHAN CS

Backend Software Development Engineer proficient in Python, Flask, REST API design, PostgreSQL, and Redis.

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PROFESSIONAL SUMMARY

Backend engineer who redesigned a browser automation system four times to scale from 1 to 150 concurrent accounts on a single EC2 instance — no infrastructure scaling, purely through architectural improvements. Built a production Flask backend sustaining 300–500 concurrent users at sub-200ms latency. Dual-degree student (IIT Madras + SLIET), with strong foundations in data structures, algorithms, object-oriented design, and scalable, fault-tolerant distributed systems.

EDUCATIONS

Indian Institute of Technology Madras Chennai, India <i>Bachelor of Science in Data Science and Applications</i>	Apr 2023 - Present CGPA : 7.36
Sant Longowal Institute of Engineering and Technology Punjab, India <i>Bachelor of Engineering in Mechanical Engineering</i>	Nov 2022 - May 2026 CGPA : 7.40

PROJECTS

Text-to-Learn AI Course Generator <i>Backend / Full-Stack Developer</i>	Mar 2026 - May 2026
- Developed a full-stack platform that transforms user prompts into structured courses, featuring modules, lessons, MCQs, and code blocks, utilizing a multi-stage Gemini API prompt pipeline. - Engineered a hierarchical MongoDB schema (Course → Module → Lesson) with Mongoose, creating over 10 RESTful endpoints for efficient CRUD operations within the content hierarchy. - Created a dynamic lesson renderer in React that supports five content block types, including syntax-highlighted code, embedded video queries, and interactive MCQs with answer explanations. - Secured user-specific routes using Auth0 OAuth2, enabling persistent course storage and individual progress tracking across sessions. <i>Technologies / Tools Used : React, Node.js, Express, MongoDB, Gemini API, Auth0</i>	
ParkSpace v2 🐙 Github <i>Backend / Full-Stack Developer</i>	Oct 2025 - Dec 2025
- Architected RBAC-enabled, scalable and fault-tolerant backend system sustaining 300–500 concurrent users at sub-200ms latency. Engineered 15+ REST API endpoints with resource-based routing and structured error handling, serving a Vue.js frontend at sub-200ms average response time. - Automated async processing of CSV exports, scheduled emails, and background reports via Celery + Redis, cutting task latency by 65% and eliminating blocking ops. - Cut endpoint response time from ~850ms to ~210ms via Redis caching on PostgreSQL dashboard queries, reducing repeated database hits by 40%. - Modeled a normalized 6-table relational schema with referential integrity and dynamic pricing rules, sustaining zero data inconsistency under concurrent load. <i>Technologies / Tools Used : Python, Flask, PostgreSQL, Redis, Celery, Vue.js</i>	
Chegg Automation System <i>Developer</i>	Mar 2024 - Dec 2024
- Engineered a browser automation system to eliminate manual 10-15 second refresh cycles, integrating Telegram API for real-time question alerts and completely removing screen monitoring. - Iterated through 4 architectural versions, scaling from 1 to 150+ concurrent accounts, diagnosing and resolving memory constraints caused by persistent browser contexts at each threshold. - Redesigned core architecture around cookie rotation model — single event loop cycling across session-per-account files — scaling to 150 concurrent accounts on one EC2 instance with no memory blowup, forming a lightweight distributed system architecture. - Implemented automatic session recovery, including dead cookie detection, re-login, and fresh cookie injection without human intervention, achieving uninterrupted 24/7 operation. <i>Technologies / Tools Used : Python, Selenium, Pyppeteer, AWS EC2, Telegram API</i>	

SKILLS

Programming Languages :	C++, Java, Python, JavaScript, SQL, C
Computer Science Fundamentals :	Data Structures, Algorithms, Object-Oriented Design, Complexity Analysis
Systems & Backend :	Distributed Systems, REST API Design, Async Task Processing, RBAC, Session Management, Cookie Rotation, Load Testing (Locust)
Databases :	PostgreSQL, SQLite, Redis, MongoDB
Frameworks & Libraries :	Flask, Vue.js, Bootstrap, Jinja2, SQLAlchemy, Selenium, Pyppeteer, Celery
Tools & Platforms :	Git, Linux, AWS EC2, Telegram API

AWARDS & ACHIEVEMENTS

- LeetCode Rating: 1857 — Top 5% globally (500,000+ users)**
- Ranked 245th** out of **30,000+** participants in LeetCode Weekly Contest 497